Cyber security roadmap

1. 🐧 Linux Mastery

Note: pwn.college is better and deeper then overthewire and you could even learn just from it

Focus on understanding the operating system that powers the internet and hacking tools.

Linux Foundations Course

Link:https://youtube.com/playlist?list=PLIhvC56v63IJIujb5cyE13oLuyORZpdkL&si=NMDP7pynB7Gc9xXO

Description: A short playlist containing the essential foundations for beginners.

BASH Scripting (Foundations)

Link: https://youtube.com/playlist?list=PLIhvC56v63IKioClkSNDjW7iz-6TFvLwS&si=4NbzRvWjBshH5-lj

Description: A funny but basic intro; good for starting, but usually not enough on its own.

BASH Scripting (Advanced)

Link: https://youtu.be/Sx9zG7wa4FA?si=po0TTSdkpQnJT4wP

Description: A longer, more advanced course for a deeper understanding of how Linux works through scripting.

Practice: OverTheWire & Pwn.college

Link: https://overthewire.org/wargames/bandit/

Description: Use SSH to solve challenges. Start with ssh bandit0@bandit.labs.overthewire.org –p 2220. Solve the first 10–12 levels.

Link: https://pwn.college/linux-luminarium

Description: A step-by-step guide with videos and labs to reach a professional level in Linux.

2. 🌐 Networking

Learn how data moves and how to build secure network architectures.

Full Networking Course

Link:https://youtube.com/playlist?list=PLIhvC56v63IJVXv0GJcl9vO5Z6znCVb1P&si=713cRWGfZIL9s_Kg

Description: Follow along to learn how to build networks and answer technical configuration questions.

3. 🪟 Windows Fundamentals

Understand the OS used in almost every corporate environment.

Windows OS Course

Link: https://youtube.com/playlist?list=PLsep1uQfemNFvT57-DOjzSnGspJv8efpS&si=FbZufeb4S0fMrFMV

Description: A short and effective course covering the basics of the Windows operating system.

Practice Tip: Download Windows in VirtualBox and practice managing users and permissions.

4. 🔑 Cryptography

The science of securing information. Focus on the basics rather than deep math.

Crypto Basics

Link: https://youtu.be/c92rMrPnFSg?si=Ca0Kj8BMXaloODcE

Description: A fundamental overview of how encryption works in a security context.

5. 🐍 Python for Security

The most important language for security automation and tool building.

Main Foundation Course

Link: https://youtu.be/ix9cRaBkVe0?si=L2GZbF3wgERg1J81

Description: Covers syntax, loops, and conditions. Can be completed in roughly 5 days.

Advanced Python Resources

Link: https://www.youtube.com/@NeuralNine/

Description: applying Python to security projects.

Quick Reference & Exercises

Link: https://www.w3schools.com/

Description: Easy tutorials with built-in challenges and exercises for quick learning.

Practice Platforms

Link: https://www.hackerrank.com/ (Beginner friendly)

Link: https://www.codewars.com/ (Aim for levels 8 through 5)